Evaluating Performance of Traditional and Neural Network Machine Learning Models in Clinical Prediction of Mortality and Length of Stays

Abstract

Computational models are seen across a variety of healthcare scenarios. From image analysis to initial risk screening in labs, these models provide vital patient insights. With the enormous amount of untapped admission data, we propose that these models can also offer more immediate benefits in hospital ecosystems. By supporting logistics through on-admission predictions of mortality and length of stay, hospitals can improve allocation of short- and long-term resources. Here, we focus on synthetic dataset of admissions based on a US hospital. We trained four machine learning algorithms- an XGBoost, LSTM, transformer, and fine-tuned language model- to perform mortality classification and hospital length of stay regression. The attention-based transformer model outperformed the other neural-nets and tree-based approaches in classification, while the XGBoost model was best at regression. Performance evaluation across different races showed consistent metrics for mortality but varied for length of stay predictions. Our results indicate the untapped capacity of admissions data for reliable clinical models and highlight the importance of training and testing multiple models for prediction, regardless of performance predispositions.

Data Analytics & Statistical Machine Learning

1 Introduction

The use of computational modelling in healthcare includes early tumour detection through image analysis of X-ray, CT and MRI data [1], to the identification of blood sugar spikes in continuous glucose monitor data. These applications showcase how machine learning models can reliably extract insights from complex biomedical big data [2].

Hospitals now hold extensive longitudinal patient admission records which provide demographics, diagnoses and other visit-specific details which may be significant predictors of clinical outcome. The data is also well-structured, clean, and tabular, making it perfect for training machine learning models. This would greatly improve the economy of motion in hospitals. Forecasting patient mortality, or computing a per-admission chance of mortality pinpoints how much priority should be given to that patient, while understanding how long patients will spend in the hospital provide a measure of future capacity estimates.

Different algorithms can be suitable to train for this context, each with their strengths and limitations. For tabular formats, XGBoost is a strong performer, utilising a traditional tree-based approach. Long Short-Term Memory (LSTM), a type of recurrent neural network has shown advantage in capturing longitudinal data such as that of patient history. Transformer-like architecture can capture non-linear relationships between features.

Our database comprised of 3 synthetic clinical relational tables and span 11,828 admissions, 5000 patients and 137,636 diagnoses. We developed the machine learning models mentioned above: XGBoost, LSTM, a transformer model, and a fine-tuned DistilBERT language model. After data pre-processing, we trained each model to perform admission-level mortality (hospital_expire_flag) classification and length of stay (LoS) regression.

The primary objective of this project is to evaluate the model's performance by assessing their efficiency in predicting the clinical outcomes. In addition, we evaluated whether the patient race had any influence in their performance, to identify whether these models can be applied in diverse places.

2 Methods

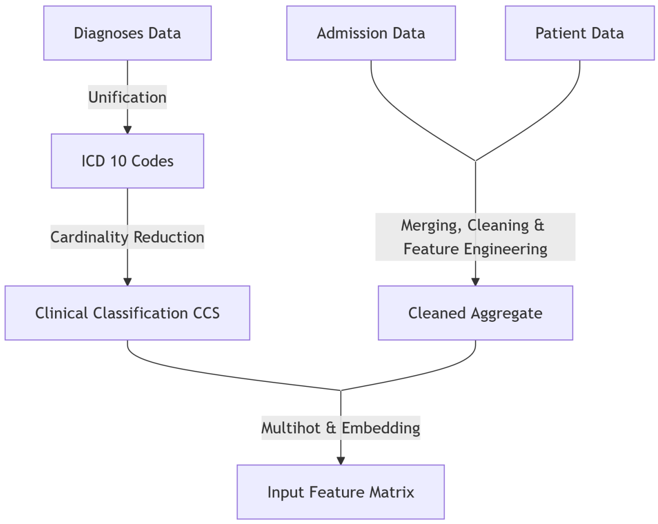

Figure 1: Flow visualisation of our data analysis workflow

2.1 Main Data-Preprocessing

Admissions data provided raw timestamps and locations for admission and discharge, as well as descriptive patient features of ethnicity, language, marital status, insurance provider. Mortality was also included, represented by a 1 in hospital_expiry_flag and 0 if survived. Another table for additional patient data stored anchor ages at an anchor year, as well as gender. These tables were merged into a single dataset. Feature engineering was performed to numerically represent dates, such as LoS, days since last discharged, age at admission.

Diagnoses data contained a mixture of ICD-9 and ICD-10 systems. ICD-9 codes were harmonised to the modern ICD-10 standard. This process required major assumptions, since there was a non-surjective map of one general ICD-9 to a greater number of more specified ICD-10 codes. Thousands of different ICD-10 diagnoses would be impractical for training, therefore a reduction in cardinality involved clustering them into a subset of 465 Clinical Classifications Software codes.

To transform diagnoses into admission-level, as well as numerically present categorical features, multi-hot encoding was performed, before joining all three tables. This process removed 10 admissions lacking diagnosis data. Class imbalances in our features were mitigated by binning under 1% of cases into an "Unknown/Other" category. Certain features were dropped due to high cardinality, class imbalance, label leakage and redundance. These included the raw timestamps, admission and discharge locations, as well as times and dates for either in- or out-of-hospital deaths. This led to our primary feature matrix, but minimal pre-processing was still required for our models. Finally, we partitioned this matrix into approximately 80% training and 20% testing data split, stratified by hospital_expire_flag.

2.2 XGBoost

XGBoost is a gradient boosting algorithm that focuses on building trees sequentially with each tree built to correct the errors of the previous one. This makes the model particularly strong for structured, tabular and imbalanced data. Features were kept in a tabular format rather than being converted to text. For missing values, these were not imputed to allow the model to use them during construction of the trees. Regularisation, tree depth constraints and feature sampling were used to control model complexity and prevent overfitting [3].

2.3 Long Short-Term Memory

LSTM networks use a gating mechanism that selectively retain/discard information across different timesteps, allowing previous admissions to influence later predictions [4]. However, this presents a challenge due to variable sequences and class imbalance in mortality prediction. Sequences were formed by chronologically ordering the admissions, where each timestep equals an admission. To improve learning, varying sequence lengths were padded, and class imbalance accounted for, by using weighted binary cross-entropy loss. Adam optimiser was used to train the model alongside gradient slipping for stability.

2.4 Transformer

A transformer architecture was implemented to model temporal relationships within a patient's sequence of hospital admissions. Unlike sequential LSTM models, transformers analyse all input tokens simultaneously, enabling modelling of long-range dependencies across a patient's admission history. Each admission was treated as a token represented by clinical and demographic features. The transformer used self-attention across all prior admissions to generate predictions per admission. Sequences were padded to equal lengths per batch and masked to ensure the model did not attend to padded positions [5].

2.5 Fine-tuned DistilBERT

DistilBERT is the distilled variant of Google's BERT: Bidirectional Encoder Representations from Transformers. As a bidirectional encoder-only framework, it excels at text predictions tasks such as sentiment analysis, rather than generative outputs from unidirectional, decoder-only architectures like GPT [6]. For each row, all features were converted into strings with a "feature: value" format, e.g. "age: 32". These were concatenated, delimiting by "|", into a single string per row. This string vector was converted into a HuggingFace dataset and tokenised via DistilBERT's AutoTokenizer with padding to a maximum sequence length of 256 tokens. Training involved 10 epochs, with a batch size of 16 and learning rate of 0.002.

2.6 Implementation Details

Models were implemented with Python 3.12.13, Pandas 2.2.2, NumPy 2.0.2, Scikit-learn 1.6.1. A seed of 67 was used for random number generation.

3 Results

3.1 Comparison Between Models

	XGBoost	LSTM	Transformer	LLM
ROC-AUC	0.968	0.940	0.984	0.525
Precision	0.240	0.197	0.629	0
Recall	0.782	0.727	0.709	0
F1-score	0.368	0.310	0.667	0

Table 1: Comparison of models for hospital mortality classification

	XGBoost	LSTM	Transformer	LLM
RMSE	5.54	5.85	5.87	7.24
MAE	2.72	2.88	2.54	4.01

Table 2: Comparison of models for length of stay regression

To assess the performance of each model for mortality classification, four metrics were used: ROC-AUC, precision, recall and F1-score [7]. Table 1 summarises the best performing model for each metric. All models except the LLM yielded high ROC-AUC values, with the transformer performing best (0.984). The transformer also achieved the highest precision (0.629), indicating fewer false positives. As for recall, XGBoost performed the best (0.782), highlighting its stronger performance in correctly identifying mortality. The transformer model performed the best in F1-score (0.667), reflecting the balance between precision and recall.

To assess performance for length-of-stay predictions, two metrics were used: Root mean squared error (RMSE) and mean absolute error (MAE) [8]. XGBoost had the lowest RMSE (5.54) out of all models, suggesting it produces fewer extreme prediction errors. The transformer achieved the lowest MAE (2.54), reflecting more accurate predictions on average across all patients.

3.2 Performance by Ethnicity

Subgroup analysis was done to examine predictive performance across ethnic groups, using the transformer model as it had the highest F1-score for mortality prediction. Results show generally similar ROC-AUC values across all groups. However, variability was seen in metrics such as F1-score, particularly in smaller ethnic groups. The same was done for length-of-stay performance. RMSE values ranged from 2.74 to 5.43 across most groups with the Unknown/Other category exhibiting a higher RMSE of 10.61. Similarly, MAE ranged from 1.71 to 4.05 across all groups, with the Unknown/Other category showing the largest error. These results are shown in Tables 3 and 4.

Ethnicity	Patients	ROC-AUC	F1-score
White	1596	0.96	0.55
Black	392	0.97	0.22
Unknown/Other	292	0.96	0.75
Asian	83	0.95	0.00
Latino	67	1.00	1.00

Table 3: Assessing the effect of ethnicity on hospital mortality classification

Ethnicity	Patients	RMSE	MAE
White	1596	5.43	2.64
Black	392	3.79	1.98
Unknown/Other	292	10.61	4.05
Asian	83	4.95	2.02
Latino	67	2.74	1.71

Table 4: Assessing the effect of ethnicity on length of stay regression

3.3 Model Evaluation

Scikit Learn was implemented for model metric evaluation. For mortality classification, we defined mortality as the positive class and survival as the negative. Performance metrics were analysed via the test set, ensuring evaluation was done on data not used in training. Model performance by ethnicity was evaluated post-training and involved indexing actual and predicted labels by each ethnicity and computing metrics on these respective subsets. Overall robustness was assessed by comparing training and test performance across models. The absence of significant differences suggests the models did not exhibit substantial overfitting.

4 Discussion

The best predictor for mortality ultimately determined using the F1-score, which balances precision and recall. This metric is particularly important in healthcare applications where both false negatives and false positives can have clinical consequences. High recall allows a model to correctly identify patients at risk of mortality, while high precision minimises unnecessary interventions. Among the evaluated models, the transformer demonstrated the highest F1-score, indicating the most balanced performance between identifying mortality cases and limiting false positives.

For length-of-stay predictions, performance was measured with RMSE and MAE. Although XGBoost exhibited the lowest RMSE, suggesting fewer extreme prediction errors, the transformer showed the lowest MAE, demonstrating greater average prediction accuracy. Consequently, the transformer was preferred due to its ability to provide more consistent predictions across admissions. In clinical environments, this stability may support hospital and patient management.

The superior performance of the transformer may be attributed to its ability to capture complex relationships across a patient's admission history. Patient outcomes are often by influenced by interactions between prior diagnoses, clinical information, and demographic factors rather than isolated admissions. The self-attention mechanism [5] enables the model to generate predictions with information from all previous admissions, identifying patterns that may span longer time periods [9]. This capacity to model long-range dependencies may explain why the transformer model achieved the strongest balance between precision and recall compared to other models. This aligns with previous research showing that transformer architectures often outperform recurrent neural networks in sequential modelling.

Subgroup analysis using the transformer across ethnic groups revealed some variation in predictive performance. However, these differences should be interpreted within the context of the data. Several ethnic groups contained few mortality events, which can cause instability in threshold-dependent metrics such as F1-score. In such cases, even a small number of incorrect predictions can disproportionately affect performance. In contrast, ROC-AUC remained consistently high across all groups, suggesting the model preserved its overall predictive power despite F1 variability. Therefore, observed differences may reflect sample size limitations rather than model bias. Further studies using larger and more balanced datasets are needed to investigate fairness in clinical prediction models [10].

Limitations for this study include a very imbalanced dataset, containing very few mortality cases. As a result, models struggled to predict mortality unless class weights were assigned to provide harsher penalisation for incorrect predictions [11]. Additionally, model interpretability was a key concern as while the models can predict risk, they don't easily explain the features driving predictions. This can result in physicians encountering difficulty in understanding the results, potentially limiting trust and adoption of these models in clinical settings.

This study highlights the potential of machine learning approaches for predicting clinical outcomes using hospital admission data. In particular, transformer architectures can effectively capture complex relationships within patient admission histories. Furthermore, subgroup analysis provides further insight into the consistency of model performance across ethnic groups. Together, these findings contribute to the expanding field of research into advanced machine learning models for clinical risk prediction and healthcare management [12].

5 Conclusion

Machine learning models offer significant potential to improve clinical outcome predictions from hospital admission data. Transformer models demonstrate strong capability in identifying mortality risk and hospital length of stay. These findings highlight how advanced machine learning can complement traditional biomedical analysis by enabling earlier risk identification and resource planning. Further validation and improved interpretability may allow models to contribute effectively to clinical decisions and ultimately improve patient healthcare.

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